

Mohamed DIAGNE

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Quant – Data Scientist – AI Engineer applying AI and machine learning to financial markets, risk modeling, and data engineering. One year of experience at Amundi and motivated to tackle new challenges in this domain. More at: [mohamed-diagne.com](#)

PROFESSIONAL EXPERIENCE

Jan 26-Present **AMUNDI ASSET MANAGEMENT** Paris, France
Data & AI Engineer, ESG Team (Consultant)

- Applied Machine Learning and NLP to semantically match unstructured ESG data to internal frameworks.
- Deployed large-scale web scraping and NLP to detect controversies and extract key signals from large ESG reports.
- Led the end-to-end design of a scalable ESG data platform (SQL, ETL, orchestration, QA, reporting) with custom internal tools.

Jul 24-Oct 25 **AMUNDI ASSET MANAGEMENT** Paris, France
Investment Risk Data Scientist, Risk Oversight

- Built *Credit Risk Intelligence*, a production-ready platform for real-time credit sentiment and AI-driven SQL queries for 300+ analysts.
- Conducted quantitative research combining econometrics and ML to predict CDS spread variations from news sentiment.
- Built end-to-end data pipelines (ETL, QA, reporting) automating data integration and powering interactive risk dashboards.
- Re-engineered the data architecture by migrating the SQL system to an SSAS Tabular cube for Power BI Cloud.

Jul 22-Aug 22 **INVICTUS CAPITAL & FINANCE SA – ASSET MANAGEMENT** Dakar, Senegal
Financial Analyst, Trading Department

- Performed fundamental analysis and issuer monitoring to support trading decisions.
 - Produced daily market notes covering sector trends and short-term trading signals.
 - Automated data pipelines (ETL, normalization, quality checks) to streamline analysis and reporting.
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EDUCATION

2023-2025 **PARIS 1 PANTHEON-SORBONNE UNIVERSITY** Paris, France
MSc. in Finance Technology & Data (M1+M2)

Relevant coursework Machine Learning, Deep Learning, Econometrics, Derivatives, Quantitative Finance, Mathematics

Thesis: Real-Time Credit Risk Prediction of Amundi's Portfolio Holdings Using News Sentiment Analysis, NLP & AI.

2020-2023 **PARIS 1 PANTHEON-SORBONNE UNIVERSITY** Paris, France
BSc. in Economics

2020 **INSTITUTION SAINTE JEANNE D'ARC** Dakar, Senegal
Baccalauréat, Economics & Social Sciences, with Honors

KEY PROJECTS [\(More available on GitHub\)](#)

PORTFOLIO OPTIMIZATION, RISK MODELING & MACROECONOMIC STRESS TESTING

- Modeled portfolio risk using Markowitz allocation and VaR/CVaR (Historical, Student-t, Monte Carlo), benchmarked to the CAC 40.
- Performed Kupiec backtesting and macro stress tests to assess portfolio sensitivity and support risk reporting.

CREDIT RISK MACHINE LEARNING MODELING & EXPLAINABLE AI

- Built end-to-end Machine Learning pipelines to predict customer default probability from financial and behavioral data.
 - Delivered role-based interactive dashboards with SHAP, surrogate trees, Anchors, and DICE to provide explainable insights to analysts, credit officers, and clients.
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SKILLS

- **Languages:** French (native), English (fluent), German (intermediate)
 - **Programming:** Python, R, SQL, VBA, Git, LaTeX
 - **Data Engineering & Automation:** Dagster, Supabase, N8N, Django / FastAPI, ETL pipelines, Web scraping, API integration
 - **Data Visualization & Tools:** Power BI (DAX), SSAS Tabular, Report Builder, Bloomberg, Jira
 - **NLP & LLMs:** Preprocessing, embeddings (OpenAI, SBERT), FAISS, RAG pipelines, reranking, RAGAS, Sentiment Analysis
 - **Machine Learning:** Logistic regression, decision trees, SVM, KNN, Bayesian, Neural networks, Cross-validation, SHAP
 - **Financial Engineering & Risk:** Black-Scholes, Greeks, CVA/DVA, VaR (parametric, historical, Monte Carlo), ES, PD/LGD/EAD, SA-CCR
 - **Statistical Modeling:** Stationarity, linear regression, ARMA, GARCH, VAR, unit roots, copulas
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ACTIVITIES & INTERESTS

- **Certification:** FRM Level 1 – GARP 2025, scholarship awarded (In Progress)
- **Finalist (Top 5/80) – Cybersecurity Challenge**, Sorbonne & Ministry of Interior: Built an AI model to detect malware via CFG analysis.
- **LLM Developer – VO2 Group:** Developed GreenSightAI, a RAG-based ESG assistant.
- **Interests:** Artificial Intelligence, Cinema, Painting, Investing, Entrepreneurship, Travelling, Golf (1 year), Tai Chi (1 year).